

15.3 As in Prob. 15.2, we know

$$\vec{\beta}_i = \beta \vec{e}_z, \quad \vec{\beta}_f = \beta (\sin\theta_p \cos\phi_p \vec{e}_x + \sin\theta_p \sin\phi_p \vec{e}_y + \cos\theta_p \vec{e}_z),$$

$$\vec{n} = \sin\theta \vec{e}_x + \cos\theta \vec{e}_z, \quad \vec{e}_1 = \cos\theta \vec{e}_x - \sin\theta \vec{e}_z, \quad \vec{e}_2 = \vec{e}_y.$$

$$\text{Then, } \vec{\beta}_i \cdot \vec{n} = \beta \cos\theta, \quad \vec{\beta}_f \cdot \vec{n} = \beta (\sin\theta \sin\theta_p \cos\phi_p + \cos\theta \cos\theta_p)$$

$$\vec{e}_1^* \cdot \vec{\beta}_i = -\beta \sin\theta, \quad \vec{e}_1^* \cdot \vec{\beta}_f = \beta (\cos\theta \sin\theta_p \cos\phi_p - \sin\theta \cos\theta_p),$$

$$\vec{e}_2^* \cdot \vec{\beta}_i = 0, \quad \vec{e}_2^* \cdot \vec{\beta}_f = \beta \sin\theta_p \sin\phi_p.$$

Using Eq. (15.2), the photon cross section is

$$\frac{d^3\sigma}{d\Omega_p d(\hbar\omega) d\Omega} = \frac{R^2}{4} \frac{e^2}{4\pi^2 \hbar c} \frac{\beta^2}{\hbar\omega} \left[\left(\frac{\cos\theta \sin\theta_p \cos\phi_p - \sin\theta \cos\theta_p}{1 - \beta (\sin\theta \sin\theta_p \cos\phi_p + \cos\theta \cos\theta_p)} + \frac{\sin\theta}{1 - \beta \cos\theta} \right)^2 + \frac{\sin^2\theta_p \sin^2\phi_p}{[1 - \beta (\sin\theta \sin\theta_p \cos\phi_p + \cos\theta \cos\theta_p)]^2} \right].$$

All we need to do now is to perform the integration w.r.t. the solid angle $d\Omega_p$, with the help from Mathematica. (TODO)

The integration is impossible to do. However, notice that the $(\vec{e}_i^* \cdot \vec{\beta}_i) / (1 - \vec{n} \cdot \vec{\beta}_i)$ term does not depend on the relative direction between the incoming particle and the outgoing radiation. Therefore, its value does not change in different coordinate system. We can then choose whatever coordinate system that can make the $\vec{e}_i^* \cdot \vec{\beta}_f / (1 - \vec{n} \cdot \vec{\beta}_f)$ term easier. To this end, we set $\vec{n}$ to be in the z -direction, with

$$\vec{n} = (0, 0, 1), \quad \vec{e}_1 = (1, 0, 0), \quad \vec{e}_2 = (0, 1, 0), \quad \vec{\beta}_f = \beta (\sin\theta_p \cos\phi_p, \sin\theta_p \sin\phi_p, \cos\theta_p).$$

$$\text{Then, } \vec{e}_1^* \cdot \frac{\vec{\beta}_f}{1 - \vec{n} \cdot \vec{\beta}_f} = \frac{\beta \sin\theta_p \cos\phi_p}{1 - \beta \cos\theta_p}, \quad \vec{e}_2^* \cdot \frac{\vec{\beta}_f}{1 - \vec{n} \cdot \vec{\beta}_f} = \frac{\beta \sin\theta_p \sin\phi_p}{1 - \beta \cos\theta_p}.$$

$$\text{and } \frac{d^3\sigma}{d(\hbar\omega) d\Omega d\Omega_p} = \frac{R^2}{4} \frac{e^2}{4\pi^2 \hbar c} \frac{\beta^2}{\hbar\omega} \left[\left(\frac{\sin\theta_p \cos\phi_p}{(1 - \beta \cos\theta_p)} + \frac{\sin\theta}{1 - \beta \cos\theta} \right)^2 + \frac{\sin^2\theta_p \sin^2\phi_p}{(1 - \beta \cos\theta_p)^2} \right]$$

$$= \frac{R^2}{4} \frac{e^2}{4\pi^2 \hbar c} \frac{\beta^2}{\hbar \omega} \left[\frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} + \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^2} + \frac{2 \sin \theta_p \cos \theta_p \sin \theta}{(1 - \beta \cos \theta_p)(1 - \beta \cos \theta)} \right]$$

Integrating with respect to $d\Omega_p$, we can see that the second term is constant and will be multiplied by 4π . The $\cos \theta_p$ term in the third term will make the integral disappear. Therefore, we only need to consider the first term. To this end, notice that

$$\begin{aligned} \int_0^\pi \frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} \sin \theta_p d\theta_p &= \int_{-1}^1 \frac{1 - x^2}{(1 - \beta x)^2} dx = \frac{1}{\beta} \int_{-1}^1 (1 - x^2) d\left(\frac{1}{1 - \beta x}\right) \\ &= \frac{1}{\beta} \int_{-1}^1 \frac{2x}{1 - \beta x} dx = \frac{1}{\beta} \int_{-1}^1 \left(-\frac{2}{\beta} + \frac{2}{\beta(1 - \beta x)} \right) dx \\ &= -\frac{4}{\beta^2} - \frac{2}{\beta^3} \log(1 - \beta x) \Big|_{-1}^1 = \frac{2}{\beta^3} \log\left(\frac{1 + \beta}{1 - \beta}\right) - \frac{4}{\beta^2} \end{aligned}$$

then

$$\begin{aligned} \frac{d^2 \sigma}{d(\hbar \omega) d\Omega} &= \int \frac{d^3 \sigma}{d(\hbar \omega) d\Omega d\Omega_p} d\Omega_p = \frac{R^2}{4} \frac{e^2}{4\pi^2 \hbar c} \frac{\beta^2}{\hbar \omega} 2\pi \left[2 \frac{\sin^2 \theta}{(1 - \beta \cos \theta)^2} + \int_0^\pi \frac{\sin^2 \theta_p}{(1 - \beta \cos \theta_p)^2} \sin \theta_p d\theta_p \right] \\ &= \frac{R^2}{4\pi} \frac{e^2}{\hbar c} \frac{\beta^2}{\hbar \omega} \left[\frac{\sin^2 \theta}{(1 - \beta \cos \theta)^2} + \frac{1}{\beta^3} \log\left(\frac{1 + \beta}{1 - \beta}\right) - \frac{2}{\beta^2} \right] \end{aligned}$$

In the limit $\beta \rightarrow 0$,

$$\log\left(\frac{1 + \beta}{1 - \beta}\right) \rightarrow \left(\beta - \frac{\beta^2}{2} + \frac{\beta^3}{3} + \dots\right) - \left(-\beta - \frac{\beta^2}{2} - \frac{\beta^3}{3} - \dots\right) = 2\beta + \frac{2\beta^3}{3} + o(\beta^3)$$

and the result will reduce to that of Problem 15.2